The Æther-flow Interpretation of Relativity

Higher-Derivative Quartic Compensator Branch Restricted Asymptotic-Closure Test on the Invariant Momentum-Suppressed Same Branch

Abstract

The momentum-suppressed same-branch note settled one narrow but decisive preliminary question on the tuned exact-square compensator branch. On the invariant restriction

$$\pi_\sigma|_{t=0} = 0, \qquad \pi_\sigma \equiv 0,$$

the full completion-specific observer-side Stueckelberg source vanishes identically, so the same repaired weighted/homogeneous class yields an honest L^1_t remainder around the same exact flat observer target. The remaining theorem burden is therefore no longer another generic-data asymptotic note and no broader redesign. It is the restricted asymptotic-closure decision on that invariant tuned sub-branch itself.

This note performs exactly that follow-up. The key structural simplification is stronger than mere integrability. Once $\pi_\sigma \equiv 0$, the propagating observer-side system contains only the same low-frequency trace–longitudinal wave pair and the same high-frequency Einstein wave block already isolated on the repaired unit-lapse route. The scalar variable survives only as the passive transport field

$$(\partial_t - \mathcal{L}_\beta)\sigma = 0,$$

so its transport-dressed profile is exact rather than perturbative:

$$\mathcal{A}_\sigma(t) \equiv \mathcal{T}^\beta(0,t)\sigma(t) = \sigma|_{t=0}.$$

The observer-side nonlinear coefficient is therefore reduced to the same quadratic wave/auxiliary remainder already recorded at the L^1_t level,

$$\mathfrak{M}_{\mathrm{ms}}(t) \lesssim \varepsilon^2(1+t)^{-2},$$

with no scalar source contribution left. The Duhamel terms for the low-frequency trace–longitudinal wave pair and the high-frequency Einstein wave block then close exactly as on the earlier positive unit-lapse asymptotic note, while the corrected tuned coercive energy satisfies an integrable-growth inequality driven by the same $\mathfrak{M}_{\mathrm{ms}}$.

The restricted verdict is therefore positive. On the invariant momentum-suppressed same branch, the same tuned exact-square completion, the same unit-lapse spatial-harmonic gauge, and the same repaired weighted/homogeneous class yield a full same-class asymptotic-closure statement around the same exact flat observer target. The observer-side perturbation remains globally small and decays to that flat target in the same weighted/homogeneous norms, the wave sectors admit limiting transport-dressed radiation profiles, and the passive scalar label carries an exact transport-dressed profile rather than a new obstruction. Generic tuned-data asymptotic claims remain closed: the positive statement proved here belongs only to the invariant momentum-suppressed sub-branch.

1. Introduction

The explicit tuned-compensator branch now has its reduced $3+1$ system, its first local well-posedness and continuation theorem, its corrected $T \sim \varepsilon^{-1}$ coercive bootstrap, its first post-coercive decay/integrability obstruction note, its first renormalized same-branch asymptotic failure note on generic tuned data, and its narrower momentum-suppressed same-branch fallback note. That last manuscript changed the question again, but only in a restricted direction. The generic tuned branch remains asymptotically closed negatively by the renormalized note. The invariant momentum-suppressed sub-branch is the only presently open asymptotic sub-branch inside the existing tuned formulation.

The remaining burden is therefore precise. One should not reopen a broader generic-data claim, and one should not redesign the branch again before this restricted question is settled. Instead, keep the same tuned exact-square completion, the same unit-lapse spatial-harmonic gauge, and the same repaired weighted/homogeneous class fixed, impose only the invariant condition $\pi_\sigma|_{t=0} = 0$, and decide whether the recorded L^1_t remainder verdict upgrades to a full same-class asymptotic-closure statement around the same exact flat observer target or instead meets the first honest obstruction there.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays within that boundary. It does not reopen generic tuned-data asymptotic claims. It does not alter the tuned local/coercive theorem architecture. It does not weaken the strict ordered-flow positivity assumptions. It decides only the remaining restricted asymptotic question on the already-isolated invariant momentum-suppressed same branch.

2. Restricted Same-Branch System and Bootstrap Regime

Retain the same unit-lapse spatial-harmonic formulation and the same basic reduced notation from the tuned local/coercive and momentum-suppressed notes:

$$N \equiv 1, \qquad V^i(\gamma) = 0, \tag{1}$$

$$h_{ij} \equiv \gamma_{ij} - \delta_{ij}, \qquad \vartheta \equiv \frac{1}{2}\delta^{ij} h_{ij}, \qquad \mathbb{K}_{ij} \equiv \Sigma_{ij} + \frac{1}{3}\delta_{ij} K, \tag{2}$$

$$F_\chi \equiv K - \mathcal{N}_\chi, \qquad \Theta \equiv \mathbf{P}_{\mathrm{lo}}\vartheta, \qquad G_\chi \equiv |\nabla|^{-1}\mathbf{P}_{\mathrm{lo}} F_\chi, \qquad \mathbf{P}_{\mathrm{hi}} \equiv I - \mathbf{P}_{\mathrm{lo}}. \tag{3}$$

Let $\mathfrak{F}^{\mathrm{tc}}_N(t)$ denote the corrected tuned-compensator coercive functional already constructed in the tuned local/coercive note.

Definition 1 (Restricted momentum-suppressed branch)**.** *Work throughout on the invariant tuned same branch defined by*

$$\pi_\sigma|_{t=0} = 0, \qquad \pi_\sigma \equiv 0, \qquad (\partial_t - \mathcal{L}_\beta)\sigma = 0. \tag{4}$$

Proposition 1 (Exact restricted same-branch structure)**.** *On the branch of Definition 1:*

1. *the observer-side tuned scalar source vanishes identically,*

$$\rho_{\rm St} \equiv 0, \qquad J_i^{\rm St} \equiv 0, \qquad \Theta_K^{\rm St} \equiv 0, \qquad \Theta_{ij}^{\rm TF,St} \equiv 0; \tag{5}$$

2. *the propagating observer-side blocks are therefore only the low-frequency trace–longitudinal wave pair and the high-frequency Einstein wave sector already isolated on the repaired unit-lapse route;*

3. *the scalar variable σ survives only as a passive transport field and no longer contributes an observer-side nonlinear source.*

Proof. The momentum-suppressed same-branch note proved (5) exactly from the current-conservation law for the tuned scalar momentum density. After that source is removed, the observer-side reduced equations are exactly the tuned Einstein equations with the completion-specific source deleted, while σ obeys the homogeneous transport law in (4). No new propagating scalar observer block remains. □

Definition 2 (Restricted asymptotic bootstrap interval)**.** *Fix $N \geq 6$, $0 < \varepsilon \ll 1$, and $C_* \geq 1$. An interval $[0, T]$ is said to lie in the* restricted momentum-suppressed asymptotic bootstrap regime *if:*

1. *the strict tuned auxiliary-elliptic regime from the tuned local/coercive note holds on every slice;*

2. *the corrected tuned coercive functional satisfies*

$$\sup_{0\leq t\leq T} \mathfrak{F}_N^{\rm tc}(t) \leq 4\varepsilon^2; \tag{6}$$

3. *the decay norms*

$$\mathfrak{D}_{\rm TL}(t) \equiv \|\nabla\Theta(t)\|_{W^{1,\infty}} + \|G_\chi(t)\|_{W^{1,\infty}}, \tag{7}$$
$$\mathfrak{D}_{\rm Ein,hi}(t) \equiv \|\partial\mathbf{P}_{\rm hi}h(t)\|_{W^{2,\infty}} + \|\mathbf{P}_{\rm hi}\Sigma(t)\|_{W^{1,\infty}} + \|\mathbf{P}_{\rm hi}K(t)\|_{W^{1,\infty}}, \tag{8}$$
$$\mathfrak{D}_{\rm aux}(t) \equiv \|\beta(t)\|_{W^{2,\infty}} + \|Q(t)\|_{W^{2,\infty}} + \|\nabla\chi(t)\|_{W^{1,\infty}} + \|W^T(t)\|_{W^{2,\infty}} \tag{9}$$

obey

$$\mathfrak{D}_{\rm TL}(t) \leq 4C_*\varepsilon(1+t)^{-1}, \tag{10}$$
$$\mathfrak{D}_{\rm Ein,hi}(t) \leq 4C_*\varepsilon(1+t)^{-1}, \tag{11}$$
$$\mathfrak{D}_{\rm aux}(t) \leq 4C_*\varepsilon(1+t)^{-1} \tag{12}$$

for all $t \in [0, T]$;

4. the low-order dispersive seed size

$$\mathfrak{S}_0^{\mathrm{ms}} \equiv \sum_{\pm} \|U_{\mathrm{TL}}^{\pm}(0)\|_{W^{2,1}\cap H^{N-2}} + \sum_{\pm} \|U_{\mathrm{Ein}}^{\pm}(0)\|_{W^{2,1}\cap H^{N-2}} \le \varepsilon \tag{13}$$

holds for the profile variables defined below.

Remark 1. *Definition 2 deliberately contains no scalar decay norm. On the invariant branch the scalar channel is not dispersive and not source-generating. It is exactly the passive transport field σ, so the restricted asymptotic question is whether the wave/auxiliary sectors close globally while that passive label remains controlled in the same closed coercive energy.*

Definition 3 (Restricted quadratic nonlinear coefficient)**.** *Define*

$$\mathfrak{M}_{\mathrm{ms}}(t) \equiv \mathfrak{D}_{\mathrm{TL}}(t)^2 + \mathfrak{D}_{\mathrm{Ein,hi}}(t)^2 + \mathfrak{D}_{\mathrm{aux}}(t)\Big(\mathfrak{D}_{\mathrm{TL}}(t) + \mathfrak{D}_{\mathrm{Ein,hi}}(t)\Big). \tag{14}$$

Proposition 2 (Restricted auxiliary decay inheritance)**.** *There exists $C_N > 0$ such that every solution in the restricted bootstrap regime satisfies*

$$\mathfrak{D}_{\mathrm{aux}}(t) \le C_N\Big(\mathfrak{D}_{\mathrm{TL}}(t) + \mathfrak{D}_{\mathrm{Ein,hi}}(t)\Big) \tag{15}$$

for all $t \in [0, T]$.

Proof. The shift, Q-sector, longitudinal solve, and transverse ordered-flow solve are the same slice equations already used in the repaired weighted/homogeneous route. On the present branch the completion-specific observer source is zero by Proposition 1, and the surviving scalar field σ enters only through the already-controlled closed coercive energy rather than through a new dispersive source term. The same weighted/homogeneous elliptic estimates therefore recover the auxiliary decay from the decaying trace–longitudinal and high-frequency Einstein sectors, giving (15). □

Corollary 1 (Time-integrability of the restricted nonlinear coefficient)**.** *There exists $C_N > 0$ such that on every restricted bootstrap interval,*

$$\mathfrak{M}_{\mathrm{ms}}(t) \le C_N C_*^2 \varepsilon^2 (1+t)^{-2}, \qquad \int_0^T \mathfrak{M}_{\mathrm{ms}}(s)\, ds \le C_N C_*^2 \varepsilon^2. \tag{16}$$

Proof. Combine (10)–(12) with Proposition 2. Every term in (14) is then $O(\varepsilon^2(1+t)^{-2})$, which is L_t^1. □

3. Transport-Dressed Principal Blocks

Define the same wave profile variables as on the earlier positive unit-lapse asymptotic-closure note,

$$U_{\mathrm{TL}}^{\pm} \equiv -|\nabla| G_\chi \pm i|\nabla|\Theta, \tag{17}$$

$$U_{\mathrm{Ein}}^{\pm} \equiv -2\mathbf{P}_{\mathrm{hi}}\mathbb{K} \pm i|\nabla|\mathbf{P}_{\mathrm{hi}} h. \tag{18}$$

Proposition 3 (Restricted principal equations)**.** *On every restricted bootstrap interval,*

$$(\partial_t - \mathcal{L}_\beta)U^{\pm}_{\mathrm{TL}} = \pm i|\nabla|\, U^{\pm}_{\mathrm{TL}} + \mathcal{N}^{\pm}_{\mathrm{TL}}, \tag{19}$$

$$(\partial_t - \mathcal{L}_\beta)U^{\pm}_{\mathrm{Ein}} = \pm i|\nabla|\, U^{\pm}_{\mathrm{Ein}} + \mathcal{N}^{\pm}_{\mathrm{Ein}}, \tag{20}$$

$$(\partial_t - \mathcal{L}_\beta)\sigma = 0. \tag{21}$$

Moreover,

$$\sum_{\pm} \|\mathcal{N}^{\pm}_{\mathrm{TL}}(t)\|_{H^{N-3}} + \sum_{\pm} \|\mathcal{N}^{\pm}_{\mathrm{Ein}}(t)\|_{H^{N-3}} \le C_N\, \mathfrak{M}_{\mathrm{ms}}(t) \tag{22}$$

for all $t \in [0, T]$.

Proof. The low-frequency trace–longitudinal and high-frequency Einstein principal equations are exactly the ones already isolated on the repaired unit-lapse route. Proposition 1 removes the completion-specific scalar observer source entirely, so the remaining remainders are only the same variable-coefficient Ricci, transport, and auxiliary-substitution terms recorded in the momentum-suppressed note. Those are precisely the terms controlled by $\mathfrak{M}_{\mathrm{ms}}$. The scalar equation (21) is the exact invariant transport law on the restricted branch. □

Definition 4 (Transport-dressed evolution families)**.** *Let*

$$\mathcal{U}^{\beta}_{\mathrm{TL},\pm}(t,s), \qquad \mathcal{U}^{\beta}_{\mathrm{Ein},\pm}(t,s)$$

denote the evolution families for the homogeneous wave blocks

$$(\partial_t - \mathcal{L}_\beta)U = \pm i|\nabla|U, \tag{23}$$

and let $\mathcal{T}^{\beta}(t,s)$ *denote the evolution family for the scalar transport equation*

$$(\partial_t - \mathcal{L}_\beta)f = 0. \tag{24}$$

Proposition 4 (Linear dispersive bounds for the restricted wave blocks)**.** *There exists* $C_N > 0$ *such that on every restricted bootstrap interval and for* $0 \le s \le t \le T$,

$$\|\mathcal{U}^{\beta}_{\mathrm{TL},\pm}(t,s)f\|_{W^{1,\infty}} \le C_N(1+t-s)^{-1}\|f\|_{W^{2,1}\cap H^{N-2}}, \tag{25}$$

$$\|\mathcal{U}^{\beta}_{\mathrm{Ein},\pm}(t,s)f\|_{W^{2,\infty}} \le C_N(1+t-s)^{-1}\|f\|_{W^{2,1}\cap H^{N-2}}. \tag{26}$$

Moreover $\mathcal{T}^{\beta}(t,s)$ *is bounded on* H^N *and satisfies*

$$\mathcal{T}^{\beta}(t,s)\mathcal{T}^{\beta}(s,r) = \mathcal{T}^{\beta}(t,r). \tag{27}$$

Proof. The wave estimates are the same perturbative transport-dressed half-wave bounds already used in the positive unit-lapse asymptotic-closure note. The transport family is generated by the homogeneous first-order equation (24), so (27) is the standard semigroup property. □

Proposition 5 (Duhamel representations)**.** *For every $t \in [0,T]$,*

$$U^{\pm}_{\mathrm{TL}}(t) = \mathcal{U}^{\beta}_{\mathrm{TL},\pm}(t,0)U^{\pm}_{\mathrm{TL}}(0) + \int_0^t \mathcal{U}^{\beta}_{\mathrm{TL},\pm}(t,s)\mathcal{N}^{\pm}_{\mathrm{TL}}(s)\,ds, \tag{28}$$

$$U^{\pm}_{\mathrm{Ein}}(t) = \mathcal{U}^{\beta}_{\mathrm{Ein},\pm}(t,0)U^{\pm}_{\mathrm{Ein}}(0) + \int_0^t \mathcal{U}^{\beta}_{\mathrm{Ein},\pm}(t,s)\mathcal{N}^{\pm}_{\mathrm{Ein}}(s)\,ds, \tag{29}$$

$$\sigma(t) = \mathcal{T}^{\beta}(t,0)\sigma(0). \tag{30}$$

Proof. Equations (28) and (29) are the usual variation-of-constants formulas for (19) and (20). Equation (30) is the exact solution formula for the homogeneous transport equation (21). □

4. Restricted Duhamel Closure

Proposition 6 (Restricted Duhamel bounds)**.** *There exists $C_N > 0$ such that every solution in the restricted bootstrap regime satisfies*

$$\mathfrak{D}_{\mathrm{TL}}(t) \le C_N\varepsilon(1+t)^{-1} + C_N\int_0^t (1+t-s)^{-1}\mathfrak{M}_{\mathrm{ms}}(s)\,ds, \tag{31}$$

$$\mathfrak{D}_{\mathrm{Ein,hi}}(t) \le C_N\varepsilon(1+t)^{-1} + C_N\int_0^t (1+t-s)^{-1}\mathfrak{M}_{\mathrm{ms}}(s)\,ds \tag{32}$$

for all $t \in [0,T]$.

Proof. Apply the dispersive estimates of Proposition 4 to the homogeneous and inhomogeneous terms in Proposition 5. The initial-data contributions are bounded by $C_N\mathfrak{S}_0^{\mathrm{ms}} \le C_N\varepsilon$. The remainders are controlled by (22). Recovering Θ, G_χ, $\mathbf{P}_{\mathrm{hi}}h$, $\mathbf{P}_{\mathrm{hi}}\mathbb{K}$, $\mathbf{P}_{\mathrm{hi}}\Sigma$, and $\mathbf{P}_{\mathrm{hi}}K$ from $U^{\pm}_{\mathrm{TL}}$ and $U^{\pm}_{\mathrm{Ein}}$ uses only bounded Fourier multipliers, giving (31)–(32). □

Proposition 7 (Convolution improvement)**.** *There exists $C_N > 0$ such that on every restricted bootstrap interval,*

$$\int_0^t (1+t-s)^{-1}\mathfrak{M}_{\mathrm{ms}}(s)\,ds \le C_N C_*^2\varepsilon^2(1+t)^{-1} \tag{33}$$

for all $t \in [0,T]$.

Proof. By Corollary 1,

$$\mathfrak{M}_{\mathrm{ms}}(s) \le C_N C_*^2\varepsilon^2(1+s)^{-2}.$$

Thus

$$\int_0^t (1+t-s)^{-1}\mathfrak{M}_{\mathrm{ms}}(s)\,ds \lesssim C_N C_*^2\varepsilon^2\int_0^t (1+t-s)^{-1}(1+s)^{-2}\,ds.$$

Split the integral into $[0,t/2]$ and $[t/2,t]$ exactly as in the earlier positive unit-lapse asymptotic note. The first half contributes $O((1+t)^{-1})$, and the second half contributes $O((1+t)^{-1})$ because $(1+s)^{-2} \lesssim (1+t)^{-2}$ there while $\int_0^{t/2}(1+r)^{-1}dr \lesssim 1+t$. This yields (33). □

Theorem 1 (Improved restricted decay bounds)**.** *Fix $N \ge 6$. There exist $C_* \ge 1$ and $\varepsilon_\sharp > 0$ such that if a smooth solution on $[0,T]$ lies in the restricted bootstrap regime of Definition 2 with*

$0 < \varepsilon \le \varepsilon_\sharp$, *then*

$$\mathfrak{D}_{\mathrm{TL}}(t) \le 2C_*\varepsilon(1+t)^{-1}, \tag{34}$$

$$\mathfrak{D}_{\mathrm{Ein,hi}}(t) \le 2C_*\varepsilon(1+t)^{-1}, \tag{35}$$

$$\mathfrak{D}_{\mathrm{aux}}(t) \le C_N\varepsilon(1+t)^{-1} \tag{36}$$

for all $t \in [0,T]$.

Proof. Insert Proposition 7 into Proposition 6. This gives

$$\mathfrak{D}_{\mathrm{TL}}(t) + \mathfrak{D}_{\mathrm{Ein,hi}}(t) \le C_N\varepsilon(1+t)^{-1} + C_N C_*^2\varepsilon^2(1+t)^{-1}.$$

Choose C_* so that $C_N \le C_*$, then choose $\varepsilon_\sharp$ so that $C_N C_*^2\varepsilon_\sharp \le C_*$. This yields (34) and (35). Equation (36) follows from Proposition 2. □

Remark 2. *Theorem 1 is the restricted nonlinear decay closure step. Once the tuned scalar source is removed exactly, the only Duhamel issue left is the same integrable wave/auxiliary remainder already identified at the* L^1_t *stage, and it improves the restricted decay bootstrap rather than saturating it.*

5. Global Restricted Asymptotic Closure

Proposition 8 (Restricted integrable-growth energy inequality)**.** *There exists* $C_N > 0$ *such that on every restricted bootstrap interval,*

$$\frac{d}{dt}\mathfrak{F}_N^{\mathrm{tc}}(t) \le C_N\mathfrak{M}_{\mathrm{ms}}(t)\mathfrak{F}_N^{\mathrm{tc}}(t) \tag{37}$$

for all $t \in [0,T]$.

Proof. The momentum-suppressed same-branch note already proved that the restricted observer-side remainder satisfies the bound

$$|\mathcal{R}_N^{\mathrm{ms}}(t)| \le C_N\,\mathfrak{M}_{\mathrm{ms}}(t)\,\mathfrak{F}_N^{\mathrm{tc}}(t).$$

That is exactly the differentiated remainder entering the corrected tuned coercive energy inequality on the restricted branch, so it yields (37). □

Corollary 2 (Uniform global restricted energy bound)**.** *There exists* $\varepsilon_\flat > 0$ *such that if* $0 < \varepsilon \le \varepsilon_\flat$ *and the solution lies in the restricted bootstrap regime on* $[0,T]$, *then*

$$\sup_{0\le t\le T}\mathfrak{F}_N^{\mathrm{tc}}(t) \le 2\varepsilon^2. \tag{38}$$

Proof. Integrating (37) and using Corollary 1 gives

$$\mathfrak{F}_N^{\mathrm{tc}}(t) \le \mathfrak{F}_N^{\mathrm{tc}}(0)\exp\Big(C_N\int_0^t \mathfrak{M}_{\mathrm{ms}}(s)\,ds\Big) \le \varepsilon^2\exp\Big(C_N C_*^2\varepsilon^2\Big).$$

For ε sufficiently small, the exponential factor is at most 2. □

Theorem 2 (Restricted same-class asymptotic bootstrap closes globally)**.** *Fix* $N \ge 6$. *There exist* $\varepsilon_{\mathrm{asy}} > 0$ *and* $C_* \ge 1$ *such that the following holds. Let compatible asymptotically flat initial data*

for the tuned unit-lapse spatial-harmonic system satisfy the hypotheses of the tuned local theorem together with the restricted condition $\pi_\sigma|_{t=0} = 0$ and the smallness assumptions

$$\mathfrak{F}_N^{\mathrm{tc}}(0) \le \varepsilon^2, \qquad \mathfrak{S}_0^{\mathrm{ms}} \le \varepsilon, \qquad 0 < \varepsilon \le \varepsilon_{\mathrm{asy}}. \tag{39}$$

Then the corresponding restricted solution exists for all $t \ge 0$ and satisfies

$$\sup_{t\ge 0} \mathfrak{F}_N^{\mathrm{tc}}(t) \le 2\varepsilon^2, \tag{40}$$

$$\mathfrak{D}_{\mathrm{TL}}(t) + \mathfrak{D}_{\mathrm{Ein,hi}}(t) \le 2C_*\varepsilon(1+t)^{-1}, \tag{41}$$

$$\mathfrak{D}_{\mathrm{aux}}(t) \le C_N\varepsilon(1+t)^{-1} \tag{42}$$

for all $t \ge 0$.

Proof. The tuned local theorem gives a nonempty short-time interval of existence. Let T_* be the supremum of times for which the restricted bootstrap regime holds. On every interval $[0,T] \subset [0,T_*)$, Corollary 2 improves the energy bootstrap (6), while Theorem 1 improves the decay bootstrap (10)–(12). Hence the restricted bootstrap assumptions improve strictly and therefore extend past T_* unless $T_* = \infty$.

The tuned local continuation theorem then yields global existence because the closed coercive norm remains finite and the strict positivity regime persists under smallness. This proves (40)–(42). □

Corollary 3 (No first honest restricted obstruction)**.** *At the present disciplined scope of the invariant momentum-suppressed tuned branch, the first honest obstruction does* not *appear at the restricted asymptotic-bootstrap stage.*

Proof. This is exactly the content of Theorem 2. □

6. Asymptotic Profiles and Restricted Verdict

Definition 5 (Restricted transport-dressed asymptotic profiles)**.** *Define*

$$\mathcal{A}_{\mathrm{TL}}^{\pm}(t) \equiv \mathcal{U}_{\mathrm{TL},\pm}^{\beta}(0,t)U_{\mathrm{TL}}^{\pm}(t), \tag{43}$$

$$\mathcal{A}_{\mathrm{Ein}}^{\pm}(t) \equiv \mathcal{U}_{\mathrm{Ein},\pm}^{\beta}(0,t)U_{\mathrm{Ein}}^{\pm}(t), \tag{44}$$

$$\mathcal{A}_{\sigma}(t) \equiv \mathcal{T}^{\beta}(0,t)\sigma(t). \tag{45}$$

Proposition 9 (Wave profiles converge and the scalar transport profile is exact)**.** *There exist limits*

$$\mathcal{A}_{\mathrm{TL},\infty}^{\pm}, \qquad \mathcal{A}_{\mathrm{Ein},\infty}^{\pm}$$

in H^{N-3} such that

$$\|\mathcal{A}_{\mathrm{TL}}^{\pm}(t) - \mathcal{A}_{\mathrm{TL},\infty}^{\pm}\|_{H^{N-3}} \le C_N\varepsilon^2(1+t)^{-1}, \tag{46}$$

$$\|\mathcal{A}_{\mathrm{Ein}}^{\pm}(t) - \mathcal{A}_{\mathrm{Ein},\infty}^{\pm}\|_{H^{N-3}} \le C_N\varepsilon^2(1+t)^{-1} \tag{47}$$

for all $t \ge 0$. Moreover,

$$\mathcal{A}_{\sigma}(t) \equiv \sigma(0) \tag{48}$$

for all $t \geq 0$.

Proof. Differentiate (43) and (44) in time and use Proposition 3. Since the dressed evolution families solve the homogeneous adjoint problems,

$$\partial_t \mathcal{A}^{\pm}_{\mathrm{TL}}(t) = \mathcal{U}^{\beta}_{\mathrm{TL},\pm}(0,t)\mathcal{N}^{\pm}_{\mathrm{TL}}(t), \qquad \partial_t \mathcal{A}^{\pm}_{\mathrm{Ein}}(t) = \mathcal{U}^{\beta}_{\mathrm{Ein},\pm}(0,t)\mathcal{N}^{\pm}_{\mathrm{Ein}}(t).$$

The evolution families are bounded on H^{N-3}, so (22) and Corollary 1 imply

$$\|\partial_t \mathcal{A}^{\pm}_{\mathrm{TL}}(t)\|_{H^{N-3}} + \|\partial_t \mathcal{A}^{\pm}_{\mathrm{Ein}}(t)\|_{H^{N-3}} \leq C_N \varepsilon^2 (1+t)^{-2}.$$

Hence the wave profiles are Cauchy in H^{N-3}, giving (46) and (47).

For the scalar profile, combine (45) with (30):

$$\mathcal{A}_\sigma(t) = \mathcal{T}^{\beta}(0,t)\mathcal{T}^{\beta}(t,0)\sigma(0) = \sigma(0)$$

by the semigroup property (27). This proves (48). □

Remark 3. *Proposition 9 is the precise restricted asymptotic outcome. The observer-side wave sectors acquire the same transport-dressed limiting radiation profiles as on the earlier positive unit-lapse route, while the scalar label is not a new Duhamel problem at all. It is already an exact transport-dressed profile.*

Theorem 3 (Restricted asymptotic verdict on the invariant momentum-suppressed branch)**.** *Under the hypotheses of Theorem 2, the invariant momentum-suppressed same branch obeys a full restricted same-class asymptotic-closure statement around the same exact flat observer target:*

1. *the observer-side perturbation remains globally small and decays to the exact flat observer background in the same repaired weighted/homogeneous norms;*
2. *the low-frequency trace–longitudinal wave pair and the high-frequency Einstein block admit limiting transport-dressed radiation profiles* $\mathcal{A}^{\pm}_{\mathrm{TL},\infty}$ *and* $\mathcal{A}^{\pm}_{\mathrm{Ein},\infty}$*;*
3. *the passive scalar label satisfies the exact transport-dressed profile law* $\mathcal{A}_\sigma(t) \equiv \sigma(0)$*, so it creates no new nonlinear asymptotic obstruction;*
4. *the correct asymptotic alternative is therefore a restricted mild transport-dressed profile, not a first honest obstruction;*
5. *no broader generic tuned-data asymptotic claim follows from this theorem.*

Proof. Item (1) follows from Theorem 2. Item (2) is Proposition 9. Item (3) is the exact identity (48). Because the scalar source vanishes identically on the branch and the remaining nonlinear coefficient is L^1_t by Corollary 1, no first honest restricted obstruction appears. The transport term $\mathcal{L}_\beta$ still forces the asymptotic description to be transport-dressed rather than literal free scattering in fixed coordinates, but that is a mild profile modification rather than a failure. The final item is exactly the claim boundary inherited from the generic-data negative renormalized note. □

Corollary 4 (Generic tuned-data claims remain closed)**.** *Theorem 3 does not alter the generic tuned-data asymptotic verdict. The broader tuned branch remains closed negatively by the renormalized*

same-branch asymptotic note; the positive asymptotic statement proved here belongs only to the invariant momentum-suppressed sub-branch.

Proof. The theorem assumptions include the exact invariant restriction $\pi_\sigma|_{t=0} = 0$. Outside that restricted branch the generic-data frozen-momentum source from the renormalized note reappears, so no broader claim is justified. □

7. What This Step Establishes

The present manuscript establishes the following.

1. It writes the substantive restricted asymptotic-closure follow-up required by the tracking docs rather than leaving that step only as an open task.
2. It keeps the same tuned exact-square branch, the same unit-lapse spatial-harmonic gauge, and the same repaired weighted/homogeneous class fixed while imposing only the invariant restriction $\pi_\sigma|_{t=0} = 0$.
3. It proves that, on this restricted branch, the only propagating observer-side sectors are the low-frequency trace–longitudinal wave pair and the high-frequency Einstein wave block, while σ becomes an exact passive transport profile.
4. It closes the restricted Duhamel argument globally because the nonlinear coefficient $\mathfrak{M}_{\mathrm{ms}}$ is L^1_t.
5. It upgrades the earlier positive L^1_t remainder verdict to a full restricted same-class asymptotic-closure statement around the same exact flat observer target.
6. It keeps generic tuned-data asymptotic claims closed.

It does not establish the following.

1. It does not reopen any broader generic-data asymptotic or nonlinear-stability claim on the tuned branch.
2. It does not show that the passive scalar label decays to zero in fixed spatial coordinates; its asymptotic description is transport-dressed.
3. It does not weaken the strict ordered-flow positivity assumptions $\kappa_\theta > 0$, $\kappa_\Omega > 0$.
4. It does not claim a broader first-principles derivation of Einsteinian gravity from substrate microphysics.
5. It does not change the editorial fact that the derivative-mixing and constraint-assisted branches are side branches rather than active derivational candidates.

8. Conclusion

The next substantive tuned-branch manuscript has now been written in the narrow form that the tracking layer required. On the invariant momentum-suppressed same branch, the tuned observer-side scalar source does not merely become integrable after renormalization. It disappears exactly.

What remains is a two-wave-sector observer problem with the same repaired weighted/homogeneous auxiliary package as before, plus a passive scalar label transported by the decaying unit-lapse shift.

That structure closes positively. The restricted nonlinear coefficient is L_t^1, the Duhamel terms for the low-frequency trace–longitudinal and high-frequency Einstein wave sectors improve the same decay bootstrap, the corrected tuned coercive energy remains globally small, and the resulting transport-dressed radiation profiles converge. The passive scalar profile is exact rather than obstructive.

The honest verdict is therefore no longer undecided on this restricted branch. The invariant momentum-suppressed tuned sub-branch does satisfy a full restricted same-class asymptotic-closure statement around the same exact flat observer target. What remains closed is broader generic tuned-data asymptotics: the positive theorem proved here is restricted to the invariant branch $\pi_\sigma \equiv 0$ and should not be read more broadly.